

Minimal In-Person Support as an Adjunct to Internet Obesity Treatment

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ABSTRACT

Background: Internet-based weight-loss programs appear promising in the short-term but, to date, have not been able to produce the level of weight loss seen in traditional in-person treatment; thus, novel approaches are necessary. Using a combination of interactive technology and in-person support has been beneficial in other areas of medicine. **Purpose:** The aim of this study is to compare 12-month weight-loss outcomes of an Internet-only behavioral weight-loss treatment with the same program supplemented with monthly in-person meetings. **Methods:** One hundred and twenty-three participants were randomized to an Internet-only ($n = 62$) or an Internet + in-person treatment (I+IPS; $n = 61$). All participants then participated in a 12-month behavioral weight-loss program conducted over the Internet. The groups met online weekly for the first 6 months and biweekly for the second half of the intervention. The I+IPS group had access to the same Web site as the Internet-only group but, once a month, attended an in-person meeting in place of an online chat. Assessments included body weight, program adherence, and social support measures. **Results:** An intent-to-treat analysis ($n = 123$) revealed there were no significant Group $\times$ Time differences ($p = .15$) in weight loss at either 6 (-6.8 ± 7.8 vs. -5.1 ± 4.8 , $p = .15$) or 12 months (-5.1 ± 7.1 kg vs. -3.5 ± 5.1 kg, $p = .17$, for Internet-only and I+IPS, respectively). Differences between groups for those completing all measures ($n = 77$) also revealed no significant differences at 6 months (-9.2 ± 7.0 kg vs. -6.9 ± 4.2 kg, $p = .08$) or 12 months (-8.0 ± 7.5 kg vs. -5.6 ± 5.5 kg, $p = .10$ for the Internet-only and I+IPS conditions, respectively). **Conclusions:** Supplementation of an Internet weight-loss treatment with monthly in-person meetings did not result in greater weight losses over 12 months. Dynamic, socially supportive, and interactive elements of the

Web site may have obviated the need for further interpersonal behavioral counseling.

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INTRODUCTION

Structured, behavioral weight-loss interventions have consistently been shown to achieve average losses of 10% of baseline weight, or about 9 to 10 kg over 20 to 26 weeks (1). However, relatively few of those who could benefit from weight-loss programs are participating in them, perhaps due to cost or inconvenience (2). Mail and telephone correspondence programs have shown promise as alternatives to in-person treatment, as have cable and interactive television-based interventions (3,4). As communications technology has progressed, the Internet has emerged as a promising way to reach greater numbers of individuals. Recent estimates have put the proportion of Americans with Internet access at 63% to 75% (5). A majority of Americans already use the Internet as a source of health information with up to 80% of Internet users looking up information about health topics (5). Moreover, delivering weight loss on the Web appears to offer benefits beyond its ability to reach large numbers of individuals in remote areas. The Internet can serve as a vehicle to deliver instantaneous, tailored feedback (6). The Internet serves as a forum for elements such as chat rooms and bulletin boards that may enhance the social support network that is already inherent in standard group behavior therapy. Finally, the Internet allows individuals anonymity, which may be appealing to some (7).

Currently, only two studies have evaluated the effectiveness of using the Internet to provide obesity treatment. Tate, Wing, and Winett (8) compared the weight loss of participants assigned to a 6-month Internet behavior therapy program with an Internet education program. All participants were given one face-to-face group weight loss session and access to a Web site with organized links to Internet weight loss resources. Participants in the behavior therapy group received additional behavioral procedures, including a sequence of 24 weekly behavioral lessons via e-mail, weekly online submission of self-monitoring diaries with individualized therapist feedback via e-mail, and an online bulletin board. The behavior therapy group lost significantly more weight than the education group (4.1 vs. 1.7 kg, $p < .005$) in 6 months. Tate, Jackvony, and Wing (9) compared an Internet weight loss program with a Internet weight loss plus a behavioral electronic counseling component. Those in the

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e-counseling group lost more weight (4.4 vs. 2.0 kg, $p = .04$) at 12 months. Weight losses in both of these studies were comparable to those achieved in other minimal contact interventions but were well below the 9 kg losses obtained in face-to-face programs (1). The authors concluded that the Internet appears to be a viable medium for conducting weight loss interventions but that further research would be needed to examine the effects of different types of contact and levels of support.

Previous research using interactive computer applications for diabetes, heart disease, cancer, and AIDS (10,11) have found that the computer has been an important adjunct to face-to-face communication. Gustafson et al. (10) randomly assigned AIDS patients to receive access to computer delivered information and peer support. All participants continued to maintain contact with their physicians. Over a 9-month period, those in the computer-mediated intervention had significantly improved quality of life, fewer hospitalizations, and more efficient (i.e., shorter) physician office visits. Cardiac patients given access to a computer-mediated counseling session delivered prior to a physician office visit had lower dietary fat and cholesterol intakes than individuals who just received physician counseling (11). It is not clear whether either of these systems would have been as effective if used in a stand-alone fashion. In fact, Glasgow, McKay, Botes, and Vogt (11) called for further research to examine "under what conditions interactive computer applications can be effective when used in a stand-alone fashion, or whether they are always best used as supplements to face-to-face care" (p. 469). In addition, a 2004 *British Medical Journal* review of peer-to-peer interaction on the Internet concluded that it was still not clear whether virtual communities could substitute or complement face-to-face support groups (12). With regard to obesity treatment, the most successful interventions are those that involve intensive, face-to-face contact with group members and a professionally trained behavioral therapist (13). Despite this, the Internet interventions for weight loss that have proliferated recently are exclusively stand-alone, self-help programs.

To date, stand-alone Internet obesity interventions have not been shown to achieve the levels of weight loss seen in in-person treatment. However, supplementing dynamic Web-based programs (which include interactive tools, engaging content, individualized therapist feedback, and peer support opportunities) with minimal in-person contact may present the benefits of online treatments with the efficacy of in-person therapies. Therefore, the aims of this study are to determine whether the Internet can stand alone as a vehicle to deliver behavioral obesity treatment or whether the addition of limited in-person support is more effective. In addition, we assess how utilization of different Web site components affect weight loss outcomes. We hypothesized that individuals receiving periodic in-person support would lose more weight than those participating solely over the Internet.

METHODS

Participants

Participants were recruited through newspaper advertisements and directed to an online application interface. To be eligible, participants had to be 18 years of age or older; have a body

mass index (BMI; kg/m^2) between 25 and 39.9; and have a computer with at least 64 MB RAM, a sound card, CD-ROM drive, a 350 MHz processor, and a 33 kbps connection speed. Those who were eligible based on computer screening were further evaluated by phone. Exclusion criteria included a history of major medical or psychiatric conditions; recent changes in medications known to affect weight; smoking (or having quit within 12 months); current, planned, or recent pregnancy; medical conditions that would prohibit exercise; and a schedule that would prohibit or restrict attendance at a designated time for weekly group meetings.

Design

Following the telephone screening, interested individuals were invited to an orientation where informed consent was obtained. Participants were randomized into one of two conditions: Internet support or Internet + in-person support (I+IPS). Because previous research has consistently determined that no-treatment control groups do not experience clinically significant weight loss (1), we chose to focus solely on treatment group differences. After randomization, individuals attended a second orientation, during which they were trained on how to use the study's Web site (VTrim). All individuals participated in a 12-month behavioral weight-loss program conducted over the Internet. The groups met online weekly for the first 6 months and biweekly for the second half of the intervention. The I+IPS group had access to the same Web site as the Internet-only group but, once a month, attended an in-person meeting in place of an online chat.

At baseline, 6 months, and 12 months, all participants were seen for objective weight measurements and were asked to complete diet, physical activity, and social support questionnaires. This study was conducted from February 2003 to March 2005 and was approved by the Committee on Human Research in the Behavioral Sciences at the University of Vermont.

Treatment

Internet condition. The 12-month weight-loss program focused on changing eating and exercise habits through behavior modification techniques, which were taught through a series of online lessons. Participants met weekly in small groups of 15 to 20 in an online chat room. They were asked to review the week's lesson and to submit, electronically, the corresponding homework assignment to their facilitator before the meeting. Although participants were not restricted from attending chat, they could not advance through the lessons without completing homework assignments. Once a lesson was completed, it remained available for the participants to review at any point.

Participants were prescribed a diet that consisted of 1,200 to 2,100 calories, based on baseline body weights. Calorie goals were determined by multiplying body weight by 12 to estimate current daily intake needs, and then subtracting 1,000 calories to produce a loss of 1 to 2 pounds per week. This technique for determining calorie intake has been used successfully in previous behavioral weight-loss studies (14). If participants achieved target body weights during the weight-loss period, calorie goals

were adjusted to maintain weight loss. The dietary intervention did not prescribe menu plans but rather emphasized eating a diet abundant in fruits, vegetables, and whole grains and moderate in fat, sugar, salt, and alcohol (in line with the U.S. Dietary Guidelines and the Food Guide Pyramid) while decreasing total calories consumed.

Initially, participants were instructed to expend 250 calories per week, exercising, ideally, 5 out of 7 days. Participants progressed 250 calories every 2 weeks until they reached 1,000 calories a week, which remained their exercise goal for the remainder of the study. An exercise program based on walking or stationary biking was prescribed; however, all forms of physical activity were encouraged. Participants were asked to record energy expended through exercise in an online journal.

Behavior therapy was delivered in the form of online lessons and corresponding homework assignments. Discussion at group chats focused on the lessons. Lesson topics included stimulus control, relapse prevention, problem solving, and social support. Participants were encouraged to discuss successes or challenges from the previous week. The therapist would then begin a review of the weekly lesson. The chats were very interactive, with all participants encouraged to "talk" regardless of their typing skills. The first few online meetings addressed self-monitoring and setting specific calorie and exercise goals. Participants were given access to an online database (Calorie King, Family Health Network, Costa Mesa, CA) to help them determine calorie counts of different foods. Each week, they submitted their journals electronically to their group facilitator, who reviewed the journals and responded with positive feedback and advice on lowering calories, increasing exercise, and achieving nutritional balance. These e-mails tended to be brief but were not structured or scripted—they were tailored for each individual based on their weekly self-monitoring data. Participants did, however, receive automated positive reinforcement every time a lesson was completed or a self-monitoring entry was made. During the second half of the study, participants received facilitator feedback on their self-monitoring journals biweekly.

I+IPS. Participants in this condition had access to the same Internet treatment program previously described. However, they met once a month as a group in person. All in-person meetings took the place of online chat sessions to keep the schedule of contact the same between conditions. Although it is not clear what the best ratio of in-person/Internet support should be, the schedule chosen was adopted as it seemed to maximize the possibility for the development of group cohesion and support while also maximizing the convenience the Internet had to offer (i.e., limiting travel to group meetings). In-person meetings for the I+IPS groups were lead by a different therapist than the online facilitator. Meetings included information that was presented that week online as well as a measured weight. We felt the model of having a local therapist who was different from the online therapist might serve as an efficient public health strategy. In this model, Internet support could be centralized with in-person support delivered locally.

Therapists

Web chats and in-person meetings were moderated by a registered dietitian and a master's level graduate student, both of whom were trained in behavior therapy principles and the VTrim curriculum. These facilitators were each responsible for specific small groups and provided the feedback on self-monitoring journals and homework assignments for their particular groups. In-person meetings were conducted by the facilitator who was not the primary online counselor for the group so as not to bias the participants' relationships with the online counselor.

Other Web Site Components

Optional Web site features designed to increase log ins and reinforce behavioral concepts were included. Participants were provided access to—but not required to use—these links to educational resources and interactive tools. Table 1 lists additional Web site components and how each related to key concepts of behavior modification or adherence to treatment goals.

Dependent Measures

All outcomes were assessed at baseline, 6 months, and 12 months unless stated otherwise. Weight change was the primary dependent measure. Weight was measured in street clothes, without shoes, on a calibrated scale. Height was self-reported. BMI was calculated by the formula, (weight in kg)/(height in m)². Dietary intake was measured with a self-report Block 98.2 Food Frequency Questionnaire, reviewed for completeness by the research team and analyzed for daily energy intake by Block Dietary Data Systems (Berkeley, CA). The Paffenbarger Physical Activity Questionnaire (15) was used to estimate energy expended in physical activity. Therapists recorded attendance at meetings, completed lessons, self-monitoring journal submissions, and weekly goals met (calories and physical activity). Perceptions of group and therapist support were measured at 6 and 12 months. The Perceived Social Support Scale (16) was used to assess the level of support group members perceive from other members of their peer group. The Working Alliance Inventory (17) gauged the participants' perceptions of therapist support. At baseline and 6 months, participants graded their level of computer experience and responded to a series of questions designed to assess attitudes toward computers.

Statistical Analysis

Body weight change was assessed using repeated measures analysis of covariance to control for baseline weight differences. An intent-to-treat analysis carrying baseline weight forward for noncompleters, as well as a completers analysis was performed. Baseline differences between groups were evaluated with independent samples *t* tests for continuous variables (age, BMI, weight, energy intake, and expenditure) and chi-squared analysis for categorical variables (race, education, marital status, and sex). Baseline values for computer ability and attitudes were highly skewed, therefore, a Mann-Whitney test was used to explore differences. Finally, repeated measures analysis of vari-

TABLE 1
Web Site Program Components

<i>Behavior Modification Program Component</i>	<i>Web Site Feature</i>
Education; skill building	Lessons/homework Online group chats Eating out guides Portion guide tutorial
Increase treatment compliance, attendance, log-ins	Contests (updated periodically) ^a Motivation page (updated periodically) ^b Carmen's story (updated weekly) ^c New flashes (updated weekly) Tips (updated weekly) Local events guide (updated periodically)
Facilitate/promote self-monitoring	Calorie databases Physical activity database Pantry Contests
Increase compliance with calorie goals	Menu planner Recipe database Contests
Increase compliance with exercise goals	Contests Local event guide Exercise counter
Provide positive feedback	Journal comments/E-mail from facilitator Automatic messages Progress charts Interactive tools/calculators (body mass index, etc.)
Foster sense of social support	Online group chats Bulletin boards E-mail with facilitator and peers Bios (member info + e-mail)

^aContests, with prizes, were held periodically to enhance motivation and facilitate adherence. ^bThe Motivation page contained menu plans, dining out tips, exercise strategies, and so forth, and was updated periodically to provide new information. ^cCarmen was the fictitious study mascot. Carmen had a weight loss history and life story designed to be motivating and captivating. Carmen's story was updated weekly.

ance was used to assess Group $\times$ Time changes in energy intake and expenditure and participants' perceptions of their computer abilities and attitudes toward computers.

A priori power calculations for weight loss were not completed as this trial was designed as a pilot to generate data for a larger investigation of Internet versus in-person weight loss treatments. All analyses were performed using the Statistical Package for Windows, version 11, for the Social Sciences (SPSS, Chicago, IL).

RESULTS

Baseline Characteristics

Baseline participant characteristics are presented in Table 2. The groups did not differ significantly in age, sex, education

TABLE 2
Participant Characteristics at Baseline

	<i>Internet-Only^a</i>	<i>I+IPS^b</i>
Age (years)	46.5 $\pm$ 10.7	47.1 $\pm$ 11.1
Body mass index (kg/m ²)	32.3 $\pm$ 3.9 ^c	31.0 $\pm$ 4.1 ^c
Weight (kg)	92.0 $\pm$ 15.7 ^c	86.1 $\pm$ 12.8 ^d
Calorie intake (kcal/day)	1897 $\pm$ 648	1936 $\pm$ 663
Fat intake (% fat/day)	38.2 $\pm$ 6.6	38.0 $\pm$ 6.6
Exercise (kcal/wk)	1,754 $\pm$ 1,380	1,474 $\pm$ 1,100
Gender		
Female	48 (77)	54 (89)
Male	14 (23)	7 (12)
Race		
White	61 (98)	61 (100)
Black	1 (2)	0 (0)
Education		
High school	4 (7)	4 (7)
Some college	16 (26)	9 (15)
College degree	24 (39)	26 (43)
Graduate/professional degree	18 (29)	22 (36)
Marital status		
Married/Civil union	45 (73)	47 (77)
Separated/Divorced/Widowed	11 (18)	10 (16)
Never married	6 (10)	4 (7)
Computer ability		
Novice	1 (2)	0 (0)
Basic software	0 (0)	4 (7)
Software + Internet	41 (66)	40 (66)
Hobbyist	13 (21)	13 (21)
Professional	7 (11)	4 (7)

Note. Age, body mass index, weight, calorie intake, fat intake and exercise are expressed as $M \pm SD$; gender, race, education, marital status, and computer ability are expressed as n (% within group). Numbers may not add up to 100 due to rounding.

^a $n = 62$. ^b $n = 61$. ^cDifferences approach significance, $p = .06$. ^dDifferences are significant, $*p = .02$.

level, or marital status. Participants (102 women, 21 men) had mean age of 46.8 years and an average BMI of 31.7 kg/m². They were predominantly White (99%), female (83%) and well-educated (74% had college degrees). The Internet-only group ($n = 62$) had significantly higher weights and BMIs than the I+IPS group ($n = 61$) at baseline ($p = .002$ and $.06$, respectively).

Attrition was 21%, however, 63% of participants provided data at all assessment points (baseline, 6 months, and 12 months). There was no significant difference between treatment groups in the number of participants who completed all three assessments (63% vs. 62% for Internet-only and I+IPS, respectively; $\chi^2 = 0.55$). Noncompleters were significantly heavier than completers at baseline (91.8 $\pm$ 18.2 kg vs. 87.4 $\pm$ 11.7 kg, $p = .01$). Otherwise, there were no significant differences between completers and noncompleters in baseline characteristics. Body weight change data are reported as an intent-to-treat analysis with baseline weight carried forward. Data for weight change are also presented for participants who completed all measures.

The remaining data reported is restricted to the 77 individuals who completed all measures at baseline, 6 months, and 12 months.

Computer Comfort and Ability

Mann-Whitney tests revealed no significant differences between the Internet-only and I+IPS groups in self-reported computer experience or in attitudes toward computers. The overwhelming majority of participants (98%) were comfortable using computers and the Internet. In general, they considered computers easy to learn (79%), a helpful health resource (96%), and an asset in time management (86%). There also were no significant changes over time or between groups in computer attitudes.

Body Weight

Using an intent-to-treat analysis ($n = 123$), there were no significant Group $\times$ Time differences ($p = .15$) in weight change at 6 months (-6.8 ± 7.8 vs. -5.1 ± 4.8 , $p = .15$) or 12 months (-5.1 ± 7.1 kg vs. -3.5 ± 5.1 kg, $p = 0.17$, for Internet-only and I+IPS, respectively; intent-to-treat results are presented in Figure 1). Although the magnitude of the weight losses were greater when only completers are examined ($n = 77$), the difference between conditions was still not significantly different at 6 months (-9.2 ± 7.0 kg vs. -6.9 ± 4.2 kg, $p = .08$) or at 12 months (-8.1 ± 7.5 kg vs. -5.6 ± 5.5 kg, $p = .10$, for Internet-only and I+IPS, respectively). There were however, significant time effects ($p < .01$) with all completers in both groups losing an average of 7.5 ± 6.4 kg or 8.5% of baseline weight over 6 months and 6.6 ± 6.6 kg or 7.5% of baseline weight over 12 months.

Adherence to Treatment Goals

Total attendance at group meetings did not differ between groups from zero to 6 months ($77 \pm 24\%$ vs. $76 \pm 16\%$, $p = .52$ for Internet-only and I+IPS, respectively) or 7 to 12 months ($56 \pm 32\%$ vs. $52 \pm 31\%$, $p = .53$). I+IPS group attended significantly fewer online meetings than the Internet-only group from zero to 6 months ($76\% \pm 21\%$ vs. $54 \pm 14\%$, $p = .00$), but from 7 to 12 months, the online attendance rates did not differ between groups ($58 \pm 33\%$ vs. $55 \pm 34\%$, $p = .67$). Participants in the

I+IPS group attended a significantly greater proportion of the online meetings from zero to 6 months (71% vs. 54% , $p < .001$); however, in-person and online attendances did not differ from 7 to 12 months (55% vs. 54% , $p = .99$).

There were no significant differences from zero to 6 or 7 to 12 months in the proportion of completed self-monitoring journals, on-time homework assignments, or physical activity and calorie goals met ($67\% \pm 30\%$ vs. $61\% \pm 27\%$, $p = .34$; $55\% \pm 35\%$ vs. $47\% \pm 30\%$, $p = .07$). Overall, participants decreased calorie intake (-420 kcal/day, $p < .001$) and increased the number of calories expended through physical activity ($+810$ kcal/wk, $p = .001$) over 12 months.

Social Support

At both 6 (7.4 ± 3.1 vs. 6.1 ± 2.8 , $p = .05$) and 12 months (7.6 ± 2.8 vs. 5.7 ± 3.5 , $p < .001$) the Internet-only group reported perceiving greater peer support than the I+IPS group. However, participants' perceptions of quality of relationships with their therapist were not significantly different between groups ($p = .56$; data not shown).

Web Site Utilization

Total log ins to the Web site were not significantly different between groups from zero to 6 (223 vs. 206 hits, $p = .60$) or 7 to 12 months (99 vs. 90 hits, $p = .69$, for the Internet-only and I+IPS, respectively). From zero to 6 months and 7 to 12 months, the Internet-only group utilized the BMI calculator more often than the I+IPS group (2.6 ± 3.0 vs. 1.0 ± 1.5 , $p = .004$); otherwise, there were no differences in the utilization of specific Web features.

DISCUSSION

Contrary to our hypothesis, monthly in-person therapy did not improve weight-loss outcomes of an online weight-loss program. These preliminary data show that individuals who received 12 months of a behavioral weight-loss program delivered over the Internet lost as much weight as those who received an online program supplemented with monthly in-person therapy. The findings of this study are promising in that, although Harvey-Berino, Pintauro, Buzzell, and Gold (18) showed that the Internet could help individuals maintain clinically significant weight losses (7.6 kg over 18 months), these data are the first to show that a weight-loss program delivered entirely over the Internet can produce a weight loss of approximately 7 kg over 6 months.

However, the results contradict our predicted outcomes. The published studies on Internet weight loss, to date, explain neither why participants randomized in the Internet-only program would reach the level of weight loss observed in the condition that received no in-person treatment as a supplement nor why an online program offering minimal in-person contact would, at 6 months, produce the level of weight loss achieved by more intense weekly in-person treatment (1). It would seem reasonable to assume, however, that Internet technologies and online programs have dramatically improved in years since the previous research was conducted.

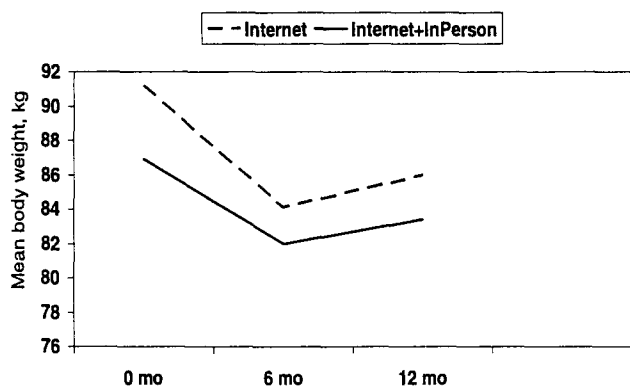


FIGURE 1 Change in body weight by condition.

In addition to individualized feedback from therapists, this study offered something that previous online weight-loss programs have not: group meetings in an online chat forum. For the Internet-only group, these meetings were conducted entirely online. For the I+IPS condition, an in-person meeting replaced the weekly Web chat once a month. The weekly meetings in this study provided not only additional participant-therapist contact but also opportunities for peer members to interact much in the same way they would at traditional in-person meetings. Although Tate et al.'s (8) research allowed participants to communicate via a bulletin board, the opportunity for "real-time" virtual interactions at chats may have fostered a greater level of perceived social support. In fact, a correlation between social support and enhanced weight loss has been reported previously (19). A sense of peer support formed in online group meetings and through the use of bulletin boards also may have promoted higher log-in frequencies. One study of an Internet-based program designed to promote behaviors conducive to diabetes management found that a group that received a peer support component on top of a standard online curriculum had higher log-in rates throughout the course of the intervention (20).

Collectively, the literature on Internet weight loss suggests frequency of Web site use is strongly correlated with weight-loss success. The results of this study support a correlation, as do Tate et al.'s (8,9) previous studies. Similarly, Christensen (21) reported that an interactive program designed to reduce the symptoms of depression was more effective for individuals who had longer-term exposure to the site. It has been argued that one of the key problems of online health interventions is that people do not use them as often as they would utilize traditional in-person treatments. Although many Internet intervention studies have not tracked log ins or Web usage, those that have reported these data suggest that compliance to online treatments decreases more rapidly than it does in in-person treatment (8,20,22). It has been proposed that improving the content of online programs in ways that prompt participants to return to the site might improve treatment outcomes (7,23). Compared with previous online weight management programs, the program delivered in this study appeared to attract a higher log-in frequency. On average, participants logged in ~200 times over 6 months, which is approximately twice the number of log ins observed in Tate et al.'s (9) 2003 (~100 over 6 months) study. Arguably, this relatively high log-in rate may have been a major contributing factor to the high levels of weight loss achieved by participants in both groups in this study. Womble's (24) findings on commercial Web site weight loss and log ins also would support this theory. Participants randomized to the online weight-loss program logged in only 17.7 times and lost a mean of 0.7 kg over 16 weeks. During the second half of the study, log ins dropped significantly ($M = 95$ during 6–12 months), which suggests that, in developing future Internet-based behavior change programs, it is necessary to focus efforts on developing tools and features that will keep people invested in long-term Internet programs.

Although it is not clear why there was a trend toward a significantly larger weight loss among participants in the Internet-only group, there may be a number of possible explanations. The Internet treatment program was designed to be identical to a standard in-person intervention, minus the in-person contact. We assumed, perhaps incorrectly, that the poor weight losses observed thus far for Internet-based programs were a result of lack of in-person group support. Because the addition of minimal in-person support did not enhance weight loss, we may have attributed more power to in-person support than it deserves. Personal contact or support may be facilitated in numerous ways and perhaps has little to do with "face time." In addition, the therapist who moderated the in-person group meeting was not the same therapist who commented on the group members' self-monitoring diaries and lessons. The therapist who facilitated the online group meetings, however, was the same therapist who read the diaries and homework assignments for the group and, subsequently, might have addressed concerns in the context of the week's lesson during the online discussion. Group members may then have felt little need for a face-to-face meeting, particularly when they were not going to be seeing their primary counselor. Finally, an in-person group meeting was but one more requirement of the research study and may have been perceived as just that—another requirement and not an opportunity for enhanced group support.

Adherence to treatment was high for both groups. Attendance has been shown to be a strong predictor of both short-term and long-term weight loss (25). Overall, participants attended 77% of all meetings during the first 6 months; this overall attendance rate is in line with those of traditional face-to-face interventions at 6 months (25,26). Participants in the I+IPS condition attended significantly fewer online (54%) versus in-person (71%) meetings at 6 months, a finding consistent with earlier findings. Assigned to the in-person treatment group, the I+IPS participants may have weighted the importance of face-to-face meetings more heavily than online meetings—although those assigned to the online-only group had to embrace the Web-based format. Attendance rates tapered significantly over the second 6 months of the study. At 12 months, the average overall and online attendances were 55% for both groups. The trend of decreased usage has been noted in other online programs (8). However, losing interest is not unique to an online delivery. Wing, Venditti, Jakicic, Polley, and Lang (27) found that during the first 6 months of in-person treatment over 2 years, attendance averaged 61% with only 27% attendance at later sessions. This tendency suggests that future research should explore methods of motivating online weight loss participants beyond the first 6 months.

As a whole, our findings underscore Tate et al.'s (9) assertion that therapist contact is a key component of a successful online weight-loss program and further suggest that, given a certain level of therapist and peer contact, online weight loss programs can facilitate clinically significant weight losses. They offered preliminary insight into how long-held theories of behavior change might be implemented into multimedia pro-

grams to enhance outcomes. However, these findings do not present a solution to the dilemma of treatment intensity or the cost associated with in-person behavioral weight-loss programs. Commercial Internet weight-loss programs that provide less intensive participant contact have a lower associated cost. However, weight losses are typically modest (28). Although the cost of this program would have been high, Internet-based programs in general can offer promise to those who do not have easy access to in-person treatment or who prefer the anonymity of Web-based meetings. Different schedules of contact or various combinations of automated and individualized feedback might reduce costs without compromising efficacy.

Future research should also examine the efficacy of larger scale (state or national) Internet weight-loss programs of different intensities to improve the generalizability of the data. Although it may not be possible to extend treatment programs as therapist-intense as the one delivered in this study to large numbers of people at no (or low) cost, it would seem feasible to reach large populations with Web sites that offer a structured behavioral curriculum, self-monitoring, and a moderate level of feedback (i.e., a combination of tailored messages and limited e-mail access or chat time with a therapist). Based on previous evidence (28), such programs might not produce the level of weight loss seen in this study but still could have a significant public health impact, given the potential "reach" of Internet-based programs.

In summary, supplementation of an Internet-only weight-loss treatment with monthly in-person meetings (in place of regular Web chats) did not result in greater weight losses. This may have been due to dynamic, socially supportive and interactive elements of the Web site that obviated the need for further interpersonal behavioral counseling.

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